

mlx cables - Cables Tool

The mlx cables tool allows users to access the cables and do the following:

- Query the cable and get its IDs
- Read specific addresses in the EEPROM
 - Read a specific register by its name. Supported registers are received by the tool (depends on the cable type)
 - Dump all the cable EEPROM bytes in RAW format
 - Upgrade the FW image on the cable uC (Only on cables that support ISSU)

mlx cables Synopsis

```
1  [-d|--dev <DeviceName>] [-h|--help] [-v|--version] [-q|--query] [--DDM] [-r|--read] [--print_raw] [--dump] [-b|--bytes_line <bytesPerLine>] [-p|--page <pageNum>] [-o|--offset <pageOffset>]
```

where:



-d --dev <DeviceName>	Perform operation for specified cable
-h --help	Show this message and exit
-v --version	Show the executable version and exit
-q --query	Query cable info
--DDM	Get cable DDM query
-r --read	Read from cable
--print_raw	Print bytes in raw format
--dump	Dump all cable pages in RAW format
-b --bytes_line <bytesPerLine>	Bytes per line in the raw print (multiples of 4, default: 4)
-p --page <pageNum>	Specific Page number to do the read/write operation
-o --offset <pageOffset>	Specific Page offset

-l --length <length>	Length of the needed data in bytes to read (default: 1 Byte)
-a --address <address>	Address (Replacement for page+offset)
--read_reg <Register>	Read register from cable
--read_all_regs	Read all registers from cable
--show_all_regs	Show all registers in the cable
--customization <customization_type>	Show cable specific customization

Notes:

- For QSFP transceivers, the tool reads the address from I2C address of 0x50. For further information, please see spec SFF8636.
- For SFP transceivers, the tool reads from I2C address 0x50 and names it page 0. When reading from I2C address 0x51 the pages will be read as page <x+1>, for example:
 - I2C address 0x51 page 0 will be referred in the tool as page 1.
 - I2C address 0x51 page 1 will be referred in the tool as page 2. For further information, please see spec SFF8472.

Examples:

To read specific byte/s in the cable pages:

```
1  mlx cables -d mt4115_pciconf0_cable_0 -r -p 0 -o 165 -l 3
2  Page[0].Byte[165] = 0x00
3  Page[0].Byte[166] = 0x02
4  Page[0].Byte[167] = 0xc9
```

Another way to read from a specific page is to use the '--address <ADDR>' flag where ADDR=0x<PAGE><OFFSET>, for example to read the same bytes with -a:

```
1  mlx cables -d mt4115_pciconf0_cable_0 -r -a 0x00A5 -l 3
2  Page[0].Byte[165] = 0x00
3  Page[0].Byte[166] = 0x02
4  Page[0].Byte[167] = 0xc9
```

To read in raw format:

```
# mlx cables -d mt4115_pciconf0_cable_0 -r -p 0 -o 128 -l 12 --print_raw

128: 0d 8c 23 81
```

```
5 132: 00 00 00 00
136: 00 00 00 05
```

To control Bytes per line, use -b:

```
1 # mlx cables -d mt4115_pciconf0_cable_0 -r -p 0 -o 128 -l 12 --print_raw -b 8
2 128: 0d 8c 23 81 00 00 00 00
3 136: 00 00 00 05
```

To query the cable:

```
1  mlx cables -d mt4115_pciconf0_cable_0 -q
2  Cable name      : mt4115_pciconf0_cable_0
3  FW version      : 2.2.550
4  FW Dev ID       : 0x21
5  FW GW version   : Legacy
6  ----- Cable EEPROM -----
7  Identifier      : QSFP+ (0dh)
8  Technology      : 1550 nm DFB (50h)
9  Compliance      : 40G Active Cable (XLPII), 100G AOC or 25GAUI C2M AOC. Provid
10 Wavelength      : 1550 nm
11 OUI              : 0x0002c9
12 Vendor          : Mellanox
13 Serial number    : MT1602FT00022
14 Part number      : MFS1200-E003
15 Revision         : AB
16 Temperature      : 54 C
17 Length          : 3 m
```

Get the DDM query of the cable:

```
# mlx cables -d mt4115_pciconf0_cable_0 --DDM
Cable DDM:
-----
Temperature : 37C
Voltage      : 3.3010V
RX Power     : -0.6712dBm
TX Power     : 0.8877dBm
TX Bias      : 6.7500mA
----- Flags -----
Temperature:
  Alarm high   : 0
  Warning high : 0
  Warning low  : 0
  Alarm low    : 0
Voltage:
  Alarm high   : 0
  Warning high : 0
```

```

18 Warning low : 0
19 Alarm low : 0
20 RX/TX Power and TX Bias:
21 RX Power alarm high : 0
22 RX Power warning high: 0
23 RX Power warning low : 0
24 RX Power alarm low : 0
25 TX Power alarm high : 0
26 TX Power warning high: 0
27 TX Power warning low : 0
28 TX Power alarm low : 0
29 TX Bias alarm high : 0
30 TX Bias warning high : 0
31 TX Bias warning low : 0
32 TX Bias alarm low : 0
33 ----- Thresholds -----
34 Temperature high alarm threshold : 80C
35 Temperature high warning threshold : 70C
36 Temperature low warning threshold : 0C
37 Temperature low alarm threshold : -10C
38
39 Voltage high alarm threshold : 3.5000V
40 Voltage high warning threshold: 3.4650V
41 Voltage low warning threshold : 3.1350V
42 Voltage low alarm threshold : 3.1000V
43
44 RX Power high alarm threshold : 5.3999dBm
45 RX Power high warn threshold : 2.4000dBm
46 RX Power low warn threshold : -10.3012dBm
47 RX Power low alarm threshold : -13.3068dBm
48
49 TX Power high alarm threshold : 5.3999dBm
50 TX Power high warn threshold : 2.4000dBm
51 TX Power low warn threshold : -8.4013dBm
52 TX Power low alarm threshold : -11.4026dBm
53
54 TX Bias high alarm threshold : 8.5000mA
55 TX Bias high warn threshold : 8.0000mA
56 TX Bias low warn threshold : 6.0000mA
57 TX Bias low alarm threshold : 5.4920mA

```

To read by register name:

Get the list of the supported registers.

```

mlxables -d mt4115_pciconf0_cable_0 --show_all_regs
Available registers per page:
=====
page00_high registers:
=====
identifier |

```

```

7      ext_identifier |
8      connector_type |
9      ..
10     ..
11     ..
12     ..
13     vendor_oui      |
      ..

```

Read the register with the register name you choose (e.g. vendor_oui, identifier).

```

1  mlx cables -d mt4115_pciconf0_cable_0 --read_reg vendor_oui
2  vendor_oui = 0x0002c9
3  mlx cables -d mt4115_pciconf0_cable_0 --read_reg identifier
4  identifier = 0x0d

```

To read all the Eeprom of the cable:

```

1  mlx cables -d mt4115_pciconf0_cable_0 --read_all_regs
2  Available registers per page:
3  =====
4  page00_high registers:
5  =====
6  identifier | 0x0d
7  ext_identifier | 0x06
8  connector_type | 0x02
9  ..
10 ..
11 ..
12 vendor_oui | 0x0002c9
13 ..

```

This will print the same tables as "--show_all_regs" but with the data that was read.

To dump all the cable's pages in raw format:

```

# mlx cables -d mt4115_pciconf0_cable_0 --dump -b 16
+-----+
| Page: 0x00 , Offset: 000, Length: 0x80 |
+-----+
000: 0d 06 00 f0 00 00 00 00 00 00 00 00 00 00 00 00
016: 00 00 00 00 00 00 29 a7 00 00 80 7d 00 00 00 00
032: 00 00 37 fa 26 2a 33 68 1c 0c b3 f9 b2 76 c2 fc
048: c1 3e 00 00 00 00 00 00 00 00 00 00 00 00 00 00
064: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
080: 00 00 00 00 00 00 00 00 00 00 00 00 00 01 00 00
096: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 08 00
112: 00 10 00 00 00 00 00 00 00 00 00 00 00 00 00 00
+-----+
| Page: 0x00 , Offset: 128, Length: 0x80 |
+-----+
128: 0d cc 23 81 00 00 00 00 00 00 00 05 ff 00 00 00

```

```
17 144: 00 00 03 50 4d 65 6c 6c 61 6e 6f 78 20 20 20 20
18 160: 20 20 20 20 1f 00 02 c9 4d 46 53 31 32 30 30 2d
19 176: 45 30 30 33 20 20 20 20 41 43 79 18 27 10 46 be
20 192: 18 07 f5 96 4d 54 31 36 31 33 46 54 30 30 36 39
21 208: 36 20 20 20 31 36 30 32 32 32 00 00 00 00 67 a9
22 224: 36 30 33 46 4d 41 32 30 33 47 31 34 32 38 20 20
23 240: 00 00 00 00 00 00 00 00 00 00 01 00 10 00 00 00
24 +-----+
25 | Page: 0x03 , Offset: 128, Length: 0x80 |
26 +-----+
27 128: 50 00 f6 00 46 00 00 00 00 00 00 00 00 00 00 00
28 144: 88 b8 79 18 87 5a 7a 76 00 00 00 00 00 00 00 00
29 160: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
30 176: 87 71 01 d3 43 e2 03 a5 00 00 00 00 00 00 00 00
31 192: 87 71 02 d4 43 e2 05 a5 00 00 00 00 00 00 00 00
32 208: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
33 224: a7 03 00 00 00 00 00 00 00 00 00 00 77 77 11 11
34 240: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
```

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